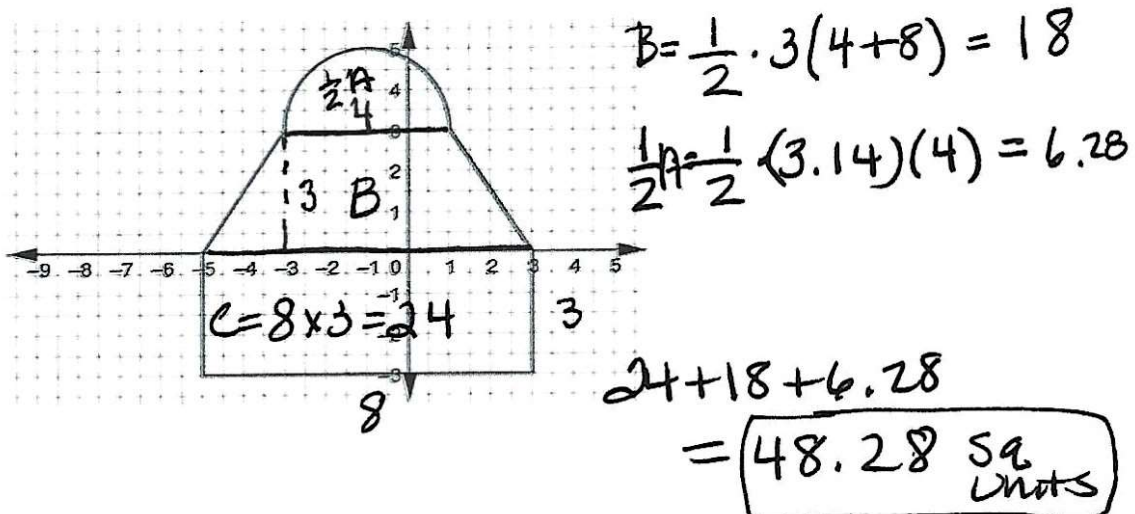
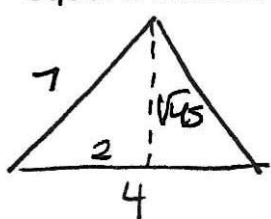


Unit 9: Worked Examples

1. Find the area.



2. A square pyramid has a slant height of 7 inches, and the area of the base is 16 square inches. What is the surface area, in square inches?



$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 4 + b^2 &= 49 \\
 b^2 &= 45 \\
 b &= \sqrt{45}
 \end{aligned}$$

$$\begin{aligned}
 &4 \left(\frac{1}{2} \cdot 4 \cdot \sqrt{45} \right) + 16 \\
 &= 53.67 + 16
 \end{aligned}$$

$$= 69.67 \text{ square inches}$$

3. The population density of a city is 2028.4 persons per square mile and the population of that town is 124,504. What is the area, in square miles, of the town?

$$\frac{124,504}{2028.4} \approx 61.38 \text{ sq miles}$$